AIX-MARSEILLE UNIVERSITY

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FACULTY OF ECONOMICS AND MANAGEMENT

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AIX MARSEILLE SCHOOL OF ECONOMICS (AMSE)

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MASTER'S DEGREE IN ECONOMETRY AND STATISTICS, DATA SCIENCE TRACK

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FINAL PROJECT

Analysis and prediction of online shoppers' purchase intent

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Introduction

The rise of online commerce has profoundly transformed purchasing behaviors and business strategies. In a digital environment where consumers have nearly unlimited access to information, understanding the determinants of purchase intention has become a central issue for both economists and e-commerce practitioners. Our study focuses specifically on analyzing and predicting online shoppers' purchase intention using behavioral data collected during their navigation on a website and downloaded from UCI Machine Learning Repository. The objective is to identify the factors that influence the probability of purchase and to develop a predictive model capable of anticipating this intention.

Indeed, the ability to convert a visitor into a buyer directly determines the performance of online businesses. As data economists, the relevance of this study goes beyond predictive performance alone. Online commerce faces a major challenge: shopping cart abandonment, which represents a significant loss of economic efficiency. Understanding the determinants of the purchase act enables companies to optimize their dynamic pricing strategies and targeted marketing efforts. The key is to identify weak signals such as the ratio between time spent on informational pages and product pages, bounce rates, or the value of the pages viewed in order to anticipate consumer needs and reduce friction along the customer journey.

The literature in digital economics and behavioral marketing has extensively examined the determinants of online purchase intention, showing that browsing behaviors constitute important predictive signals. Several foundational studies highlight that online purchase decisions are influenced by information quality, search costs, and user experience.

Bakos (1997) modeled the role of search costs in electronic markets. His analysis shows that reducing information costs for buyers decreases sellers' ability to extract monopolistic profits and improves resource allocation in differentiated product markets. This foundational work establishes the theoretical basis of e-commerce platform economics.

Building on these theoretical foundations, recent empirical research has focused on identifying the contextual and behavioral signals that influence purchase intention. **Xiao et al. (2019)** analyzed the influence of contextual cues on purchase intention in cross-border e-commerce. Based on a survey of 372 Chinese consumers and a PLS-SEM model, their results show that four signals—online promotions, content marketing, personalized recommendations, and user reviews—have a significant positive impact on purchase intention, while brand awareness exerts a negative moderating effect. In the same vein, **Sakar et al. (2019)** compared Random Forest, SVM, and Multilayer Perceptron (MLP) on the *Online Shoppers Purchasing Intention* dataset, showing that MLP provides the best precision and F1-Score performance for predicting online purchase intention.

More recently, attention has shifted toward the analysis of clickstream data and algorithmic optimization. **Wang and Zhang (2023)** examined the prediction of purchase intention using clickstream data with Random Forest and Gradient Boosting. Their results show that engagement variables-such as add-to-cart events, time spent on product pages, number of pages viewed, and visit history—are the most discriminating factors, while demographic variables only marginally improve predictive performance. **Swetha et al. (2024)** compared seven classification algorithms—Random Forest, Decision Tree, SVM, Gradient Boosting, AdaBoost, Light-

GBM, and CatBoost on the *Online Shoppers Purchasing Intention* dataset. Their results show that CatBoost, with hyperparameter optimization, achieves the highest accuracy (98.73%) for predicting online purchase intention.

Finally, the most recent studies incorporate explainability approaches to improve the interpretability of predictions. **Jayanthi et al. (2025)** explored e-commerce revenue prediction by comparing eight algorithms XGBoost, Random Forest, Logistic Regression, SVM, Decision Tree, KNN, Gradient Boosting, and Voting Classifier on the *Online Shoppers Purchasing Intention* dataset. With SMOTE and hyperparameter optimization using GridSearchCV, Random Forest achieves the best performance (Accuracy: 92.45%, F1-Score: 92.59%, ROC-AUC: 97.88%). The use of SHAP (XAI) makes it possible to identify the key variables influencing conversion.

These studies reveal remarkable scientific progress, moving from theoretical foundations to high-precision operational predictive capabilities. Their value lies in a dual contribution: transforming behavioral traces into concrete levers for optimizing purchase journeys in real time, while reconciling predictive performance with the explainability of decision-making mechanisms. In a context of increasing competition and rising consumer expectations, mastering these approaches has become a key factor of strategic differentiation.

To address this issue, we adopt an empirical approach based on a real dataset describing navigation sessions on an e-commerce website. Our methodology is structured in several steps. We first conduct an exploratory analysis to characterize visitor behaviors and identify potentially influential variables. We then apply dimensionality-reduction techniques (PCA) and variable-selection methods to simplify the information space while retaining the most relevant dimensions. Given the imbalance in our dependent variable “Revenue” between sessions with and without purchase, we use the SMOTE technique (Synthetic Minority Over-sampling Technique) to rebalance the classes and prevent models from systematically favoring the majority class. Finally, we build several predictive models (logistic regression, random forest) to assess the ability of behavioral data to anticipate purchase intention.

The descriptive analysis shows that only 15.6% of visits result in a purchase, confirming a strong imbalance between the classes. Moreover, the results indicate that a new visitor has a higher probability of purchasing than other types of visitors. To model this behavior, three algorithms were used: logistic regression, Random Forest, and KNN. Based on the AUC, a metric particularly well suited to imbalanced data, the best-performing model is logistic regression, with an AUC of 89.7%. The extraction of coefficients followed by the computation of marginal effects then makes it possible to interpret the impact of the principal components on purchase probability. Each marginal effect represents the change in this probability (in probability units) induced by a one-unit increase in a principal component, all else being equal.

Thus, this study contributes to a better understanding of the economic mechanisms at work in online commerce and offers a robust empirical approach to analyzing and predicting purchase intention using behavioral data.

Materials and Methods

1 Data Presentation

1.1 Data Source

The data analyzed come from the Online Shoppers Purchasing Intention Dataset, widely used for studying online purchasing behavior. It contains 12,330 browsing sessions on an e-commerce website, each described by 18 explanatory variables and a binary target variable (Revenue), indicating whether the visit resulted in a purchase or not.

1.2 Variable Description

The variables are divided into several categories:

Variables	**Descriptions**	**Types**
Administrative	Number of administrative pages visited	Continuous numeric
Administrative_Duration	Time spent on administrative pages	Continuous numeric
Informational	Number of informational pages visited	Continuous numeric
Informational_Duration	Time spent on informational pages	Continuous numeric
ProductRelated	Number of product pages visited	Continuous numeric
ProductRelated_Duration	Time spent on product pages	Continuous numeric
PageValues	Average value of visited pages	Continuous numeric
SpecialDay	Proximity to a special day	Continuous numeric
BounceRates	Bounce rate	Proportional numeric
ExitRates	Exit rate	Proportional numeric
Month	Month of visit	Categorical
OperatingSystems	Operating system used	Categorical
Browser	Browser used	Categorical
Region	Visitor's geographic region	Categorical
TrafficType	Traffic type	Categorical
VisitorType	Visitor type (new, returning, other)	Categorical
Weekend	Visit made on a weekend (Yes/No)	Categorical
Revenue	Purchase intention (0 = no purchase, 1 = purchase)	Target variable

2 Exploratory Analysis

After importing the dataset, a verification revealed 125 duplicates, representing 1.01% of the initial sample. To ensure the quality of the analyses, these observations were removed. The final dataset therefore contains 12,205 unique sessions, providing a clean and reliable basis for the rest of the study.

2.1 Descriptive Statistics

The descriptive analysis shows strong heterogeneity in navigation durations and highly skewed distributions for temporal variables. It also highlights a low proportion of sessions that resulted in a purchase, revealing a significant class imbalance.

2.2 Correlation Analysis

The analysis of the correlation matrix highlights structural relationships between the numerical variables, notably moderate correlations between navigation durations and the number of pages viewed. PageValues appears to be strongly linked to the target variable Revenue, underscoring its predictive importance.

3 Data Preparation

String-type variables, such as Month, were transformed using one-hot encoding, while the variable VisitorType was encoded using label encoding. Once these transformations were completed, standardization was applied to all variables in order to harmonize the scales before applying PCA as well as models such as logistic regression, random forest, and KNN.

4 Modeling

4.1 Dimensionality Reduction: Principal Component Analysis (PCA)

PCA is used to reduce dimensionality, eliminate collinearity, and improve model stability, while also speeding up distance-based methods such as KNN. Applied to standardized data, it allows the selection of components that explain the most variance. The first 47 components, capturing most of the information, are then used as new explanatory variables for the classification models.

4.2 Handling Class Imbalance

The analysis of the target variable shows a strong imbalance: 84% of sessions do not result in a purchase, compared to only 16% that do. This imbalance may bias the models toward the majority class, hence the need for rebalancing. To address this, the SMOTE method was used to generate synthetic observations of the minority class and improve the detection of sessions leading to a purchase. To avoid any data leakage, SMOTE was applied only to the training set after the data split.

4.3 Logistic Regression

Logistic regression is a probabilistic linear model based on the sigmoid function to estimate the probability of belonging to the positive class. It offers several advantages, including high interpretability, strong robustness, and reliable performance on previously standardized data. In this study, logistic regression was trained on the principal components obtained from PCA, after applying SMOTE to the training set to correct the pronounced class imbalance. In addition, the "balanced" option was activated to strengthen the consideration of the minority class and limit biases related to the initial data distribution. This model therefore serves as a solid benchmark for assessing the contribution of more flexible or non-linear methods.

4.4 Random Forest

Random Forest is an ensemble method based on aggregating a large number of decision trees built on random subsamples of both data and variables. This approach has the advantage of being particularly robust to noise, effectively capturing non-linear interactions between variables, and being insensitive to data scaling. In this analysis, a model composed of 300 trees was selected to ensure stable predictions. As with logistic regression, SMOTE was applied to the training set before model training, allowing the classes to be balanced and improving the detection of the minority class. Thanks to its flexibility and its ability to model complex relationships, Random Forest serves as a relevant benchmark compared with linear models and distance-based methods. It was further optimized using GridSearchCV.

4.5 K-Nearest Neighbors (KNN)

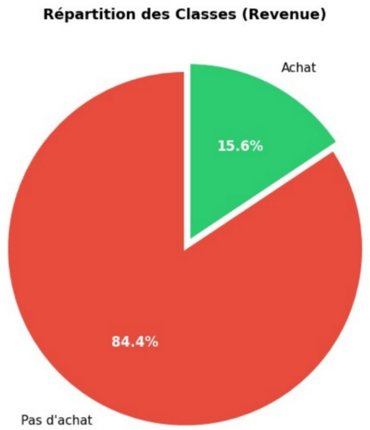
K-Nearest Neighbors is a non-parametric model based on Euclidean distance to classify an observation according to its nearest neighbors. This method requires strict data standardization, as it is particularly sensitive to scale differences between variables. Since the choice of the number of neighbors is decisive for model performance, an exhaustive search using GridSearchCV was conducted to optimize the hyperparameters: n-neighbors (between 1 and 50), weights (distance), and metric (Euclidean). The best model obtained corresponds to k=8, with Euclidean distance and distance weighting. Unlike logistic regression and Random Forest, SMOTE was not applied for KNN, as the generation of synthetic points can alter the local structure of the data and distort distances, which would degrade model performance.

Results

5 Exploratory analysis

5.1 Distribution of the Target Variable

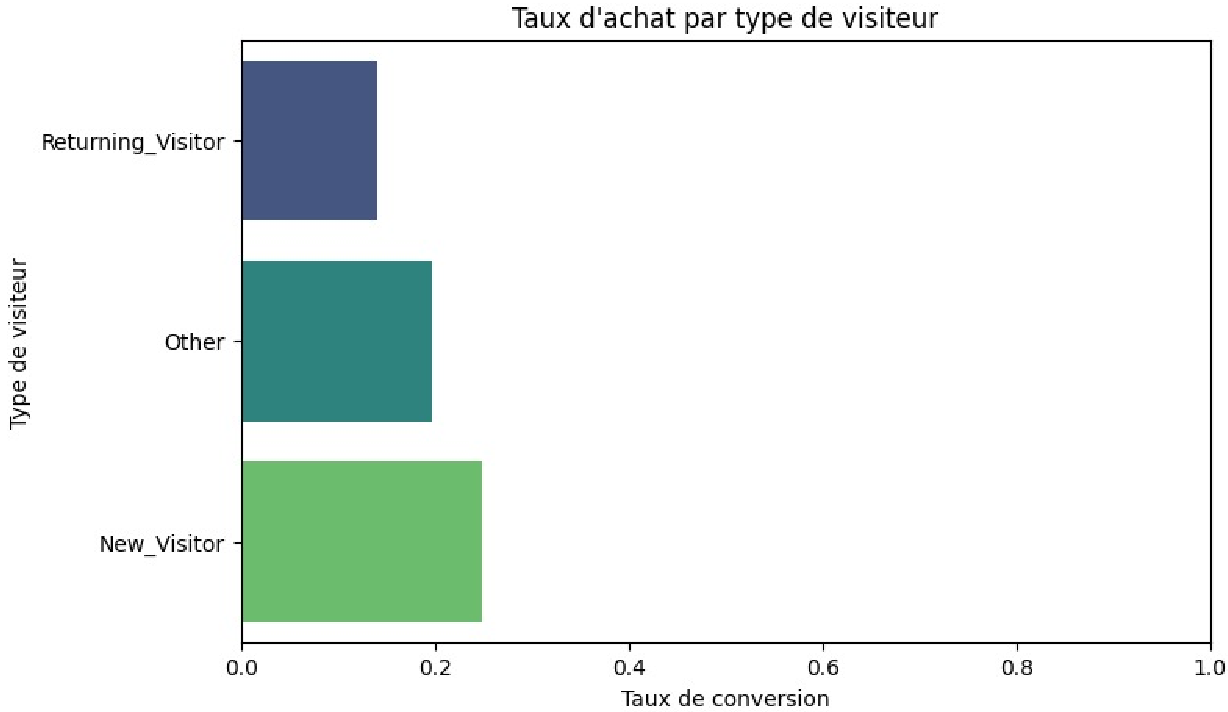
The analysis of the distribution of the target variable *Revenue* reveals a marked imbalance between the two classes. Out of the 12 205 analysises sessions, 10 310 (84,4 %) did not result in a purchase, while only 1 895 (15,6 %) resulted in a transaction.



5.2 Purchase Rate by Visitor Type

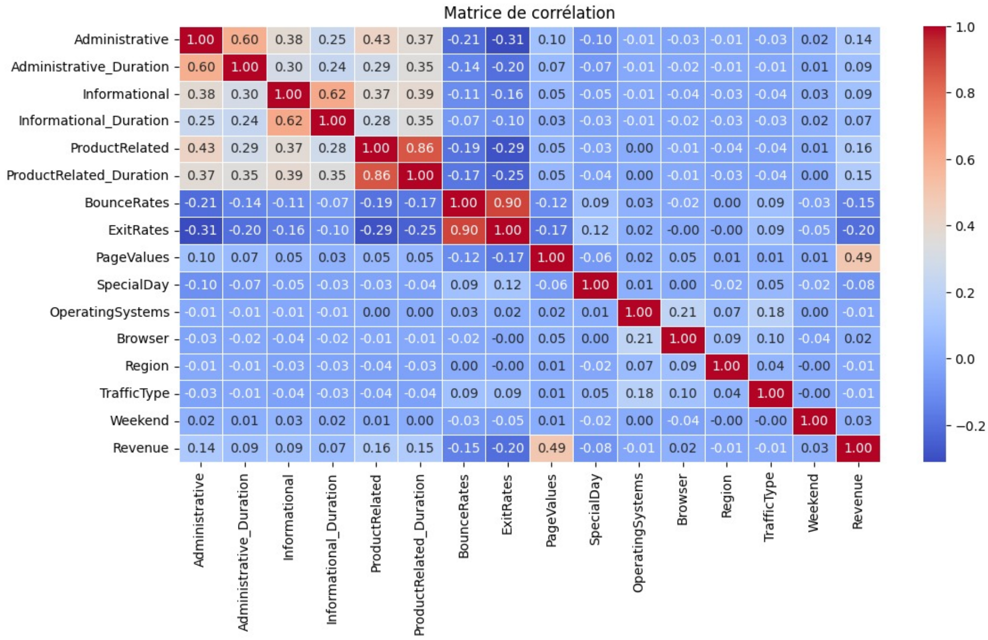
The analysis of conversion rate by visitor type highlights differentiated purchasing behaviors. New visitors (*New_ Visitor*) present a conversion rate of 24,9 %, significantly higher than that of returning visitors (*Returning_ Visitor*) which stands at 14,0 %.Visitors classified in the category *Other* meanwhile display an intermediate rate of 17,3 %.

This result can be explained by the fact that new visitors often arrive on the site with a more definite purchase intent, potentially guided by targeted advertising or a specific product search. Conversely, returning visitors may include a significant proportion of users in a comparison or browsing phase, without immediate purchase intent.



5.3 Correlation Analysis

The correlation matrix of numerical variables highlights several structuring relationships. Notable observations include:
- A very strong correlation (0.91) between *BounceRates* et *ExitRates*.
- High correlations between duration variables and the number of pages visited for each category (Administrative, Informational, ProductRelated).
- A notable positive correlation between *PageValues* et *Revenue* (0.49), confirming the predictive power of this variable.



5.4 Results of the Principal Component Analysis

The application of PCA on the standardized variables made it possible to identify the share of total variance explained by the different components.

Component	Eigen Value	Proportion (%)	Cumul (%)
PC1	3.457	13.83	13.83
PC3	1.626	6.50	27.96
PC6	1.187	4.75	43.80
PC9	1.049	4.20	56.94
PC12	0.986	3.94	69.19
PC15	0.935	3.74	80.60
PC18	0.778	3.11	90.65
PC21	0.420	1.68	97.59

To ensure 90% representativeness of the data, we retain the first 18 principal components. This choice reduces dimensionality from 25 to 18 variables while limiting information loss to less than 10 %.

In terms of Main components and dominant variables, we have:

Composante	Variable Dominante	Loading	Contribution (%)
PC1	ProductRelated_Duration	-0.406465	16.52
PC2	BounceRates	-0.424084	17.98
PC3	Month_May	0.586696	34.42
PC4	OperatingSystems	-0.530868	28.18
PC5	Month_Nov	0.670480	44.95
PC6	Month_Mar	0.596055	35.53
PC7	Month_Dec	0.624029	38.94
PC8	Weekend	0.367147	13.48
PC9	Month_Jul	-0.679695	46.20
PC10	Month_Sep	0.696599	48.53
PC11	Month_June	0.694637	48.25
PC12	Month_June	-0.520581	27.10
PC13	Weekend	0.655564	42.98
PC14	Region	-0.816800	66.72
PC15	Month_Oct	0.450047	20.25
PC16	PageValues	-0.611780	37.43
PC17	Browser	-0.639511	40.90
PC18	VisitorType	0.729569	53.23

6 Predictive analysis

6.1 Application of SMOTE and Data Distribution

The data were divided into a training set (70%) and a test set (30%), using stratification on the target variable to preserve the class distribution. The SMOTE technique was then applied exclusively to the training set to rebalance the classes

Classe	Before SMOTE	After SMOTE
Non-achat (0)	7 207 (84,4 %)	7 207 (50,0 %)
Achat (1)	1 336 (15,6 %)	7 207 (50,0 %)
Total	8 543	14 414

The test set, composed of 3 662 observations, retains the original class distribution and is used only for the final evaluation of the models.

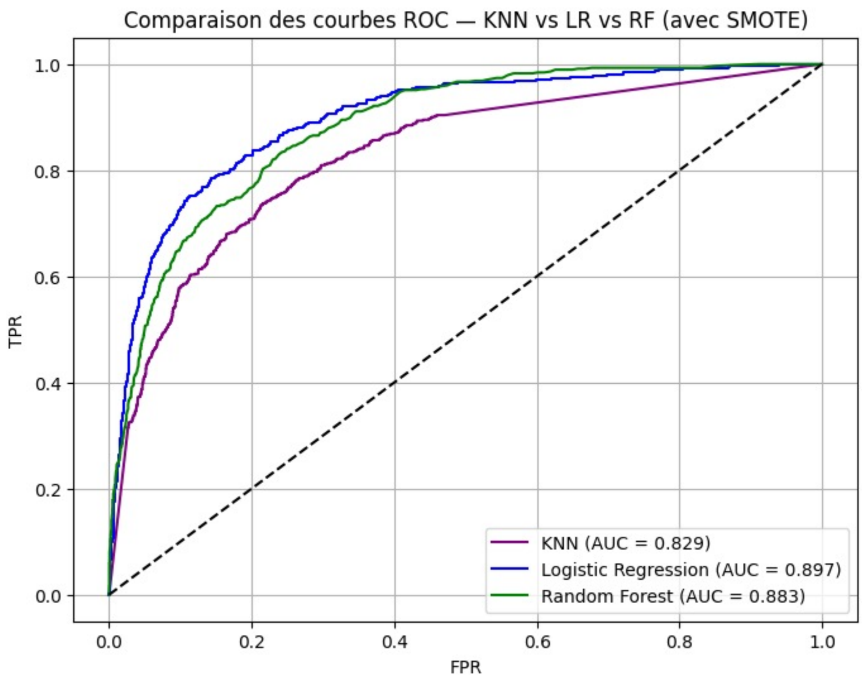
6.2 Overall Comparison of Model Performance

The three classification models were evaluated on the test set using several complementary metrics. The table below presents a summary of the performance obtained.

Model	Accuracy	Precision	Recall	F1-Score	AUC
Logistic Regression	0.86	0.89	0.86	0.87	**0.897**
Random Forest	0.87	0.87	0.87	0.87	0.883
KNN (k=8)	0.80	0.85	0.80	0.82	0.829

Although the Random Forest models (`n_estimators=300`, `max_depth=20`, `min_samples_split=5`, `min_samples_leaf=2`) et KNN (`n_neighbors=8`, `weights='distance'`, `metric='manhattan'`) were optimized using GridSearchCV, logistic regression emerges as the best model according to the AUC metric, with a value of 0.897, higher than those obtained by Random Forest (0.883) and KNN (0.829).

In the context of e-commerce, where identifying potential buyers is crucial for marketing targeting, recall is of particular importance. In this regard, logistic regression, with a recall of 0.86, makes it possible to detect a greater proportion of sessions resulting in a purchase.



6.3 Detailed Results of Logistic Regression

Logistic regression, trained on the 18 principal components derived from PCA, constitutes the selected model due to its superior AUC and interpretability. The detailed classification report is presented below.

Classe	Precision	Recall	F1-Score	Support
Non-achat (0)	0.95	0.87	0.91	3 090
Achat (1)	0.53	0.76	0.62	572
Macro avg	0.74	0.82	0.77	3 662
Weighted avg	0.89	0.86	0.87	3 662

The model correctly identifies 76% of sessions with a purchase (recall of class 1), while maintaining a precision of 53%. This trade-off is acceptable in a context where the cost of a false positive (targeting a non-buyer) is generally lower than the cost of a false negative (missing a potential buyer).

6.4 Marginal Effects

To facilitate economic interpretation, average marginal effects were calculated. Unlike odds ratios, marginal effects directly express the change in purchase probability (in percentage points) associated with a one-unit increase in each principal component.

Variable Dominante	Code	Effet Probabilité	Impact Achat (%)
PageValues	PC16	-0.1908	-19.08
Weekend (v2)	PC13	-0.1748	-17.48
BounceRates	PC2	0.1234	12.34
ProductRelated_Duration	PC1	-0.1056	-10.56
Browser	PC17	0.0823	8.23
Month_Oct	PC15	-0.0791	-7.91
Month_Nov	PC5	0.0744	7.44
OperatingSystems	PC4	-0.0689	-6.89
VisitorType	PC18	0.0579	5.79
Month_Mar	PC6	-0.0555	-5.55
Month_June (v2)	PC12	-0.0473	-4.73
Weekend	PC8	0.0402	4.02
Month_May	PC3	0.0391	3.91
Month_Sep	PC10	0.0342	3.42
Region	PC14	-0.0250	-2.50
Month_Dec	PC7	-0.0203	-2.03
Month_Jul	PC9	-0.0146	-1.46
Month_June	PC11	-0.0034	-0.34

The average marginal effects clearly highlight which dimensions most strongly influence the probability of purchase. The two most detrimental factors are PageValues and Weekend (v2), which reduce the likelihood of purchase by roughly 19% and 17%, respectively, suggesting that low-value pages or certain weekend browsing patterns significantly hinder conversion. Conversely, behaviors such as higher BounceRates, the use of specific browsers, or the visitor type slightly increase the probability of purchase, although these effects remain more modest (between +4% and +12%). Temporal variables reveal a nuanced seasonality: November, May, and September slightly boost purchase likelihood, whereas October, March, June, and December weaken it. Technical characteristics such as operating systems or region have a negative but limited impact. Overall, the model reflects a combination of behavioral, temporal, and technical effects, with page value and certain weekend-related navigation patterns standing out as particularly influential.

Conclusion

This study aimed to analyze and predict the purchase intent of e-commerce website visitors based on their browsing behaviors. The exploratory analysis revealed a strong imbalance in the target variable, justifying the use of SMOTE to rebalance the classes. The application of PCA effectively reduced dimensionality while retaining the essential information, thereby facilitating modeling. Among the models tested, logistic regression proved to be the most performant, with an AUC of 0.897, outperforming Random Forest and KNN despite their optimization. The study of marginal effects also made it possible to identify the most influential components, notably PC4, which is strongly linked to the PageValues variable.

From a managerial standpoint, the results underscore the importance of product page quality, targeting new visitors, and taking seasonality into account in marketing strategies. The selected model, both high-performing and fast, can be deployed in real time to identify sessions with high conversion potential and trigger personalized actions. These insights thus offer concrete avenues for optimizing the customer journey and improving the commercial performance of e-commerce platforms.

This study presents several limitations that open up as many avenues for future work. The dataset, derived from a single e-commerce site and lacking essential information such as prices, product characteristics, or sociodemographic data, limits generalization and a nuanced understanding of purchasing mechanisms. The interpretation of principal components also remains partial, and the selected model, although performing well in terms of AUC, does not detect all potential buyers, which may represent a significant operational challenge. To overcome these limitations, avenues for improvement include data enrichment, exploration of more advanced models such as Gradient Boosting or neural networks, integration of explainability tools such as SHAP, as well as the development of models capable of learning in real time. Finally, a causal approach would make it possible to go beyond prediction to assess the actual impact of marketing interventions on conversion.

Ultimately, this study demonstrates that behavioral browsing data constitutes a valuable source of information for anticipating online purchase intent. By combining methodological rigor with an economic perspective, our approach offers e-commerce practitioners concrete tools to optimize their conversion strategies and improve the efficiency of their marketing investments.

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